

Forest Surveying and Mapping

Land Sciences

May, 2026

Forest Surveying and Mapping (A06215)

Short Title: Forest Surveying and Mapping
Faculty Science and Computing
Department Land Sciences
Cluster Forestry
Author MPEDINI
Credits 5

Level: Introductory

Description of Module / Aims

In this module students will learn how to read a map and use it, along with a compass, for navigation in the landscape. Students will also learn how to survey (measure) a piece of land using Global Navigation Satellite System/Global Positioning Systems (GNSS/GPS) supported by classic surveying techniques. PC's and mobile devices will be used to display the spatial surveying data. Furthermore, students will experience how to collect spatial data for use with Geographic Information Systems (GIS).

Programmes

FORS-0005	BSc in Forestry (WD_SFORS_D)	1 / 2 / M
FORS-0005	BSc in Forestry (WD_SFORS_D)	2 / 4 / E

Indicative Content

- Types of maps, map scale, map-projections, map reading, cartographic symbols, Ordnance Survey Ireland (OSI) mapping products, forest maps
- Distance measurements on flats and slopes with and without obstacles; offsets, perpendiculars and parallels
- Basic linear surveying techniques using offsets and trilateration
- Measuring distances and areas off a map using scale rulers and hectare grid
- Land navigation using a compass
- Measurement of horizontal angles using a compass; resection and intersection
- Projected co-ordinates systems; co-ordinate geometry calculations
- Basic concept of Global Navigation Satellite System (GNSS), including expected accuracy
- Best practice for satellite based surveying

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify mapping symbols, and describe a landscape based on map information alone.
2. Describe geographical and projected co-ordinate systems such as WGS84, IG, ITM and UTM and convert between them using online tools.
3. Calculate grid co-ordinates using basic co-ordinate geometry formulas.
4. Use maps for navigation, localisation, distance and area measurements. This includes finding locations on a map having been given grid references or grid co-ordinates.
5. Use best working practices for recreational and mapping grade GNSS receiver to survey a sizeable area of forest or agricultural land.
6. Describe the fundamental principles of satellite based navigation systems and their practicality and limitations in land surveying.
7. Apply basic land surveying techniques in conjunction with co-ordinate geometry calculations to establish the grid co-ordinates of survey points.
8. Display surveying data in an appropriate map format along with a description of methodology and an assessment of accuracy.

Learning and Teaching Methods

- Lectures: will be used to give student the theoretical background for mapping and cartography and the theory behind GNSS.
- Field trips: will be used for practical work on GNSS surveys and traditional navigation and surveying techniques. Basic GNSS surveying. Basic land surveying techniques for co-ordinate calculations. GNSS surveying of a piece of land.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,6
Continuous Assessment	50%	
Assignment	15%	1,2,4
Assignment	35%	5,7,8

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations

when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self-study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Field Trips	24	
Independent Learning	87	

Essential Material

- "GPS Tutorial ." http://www.trimble.com/gps_tutorial/
- "Ordnance Survey Ireland." www.osi.ie
- Irvine, W. and F. MacLennan. _Surveying for construction._ . 5th Edition. New York: McGraw-Hill Book Company, 2006.

Supplementary Material

- "Map Reading." <http://www.ordnancesurvey.co.uk/docs/ebooks/map-reading.pdf>
- "The Cartographer's Craft Project." www.colorado.edu/geography/gcraft/contents.html

Requested Resources

- Room Type: Computer Lab